MICROSOFT POWER BI
PART I

INTRODUCTION

POWER BI OVERVIEW

- Power BI is a technology-driven business intelligence tool provided by Microsoft for analyzing and visualizing raw data to present actionable information.
- It combines business analytics, data visualization, and best practices that help an organization to make data-driven decisions.
- Microsoft as Leader in the

  "2024 Gartner Magic Quadrant for **Analytics**

  and Business Intelligence Platforms"



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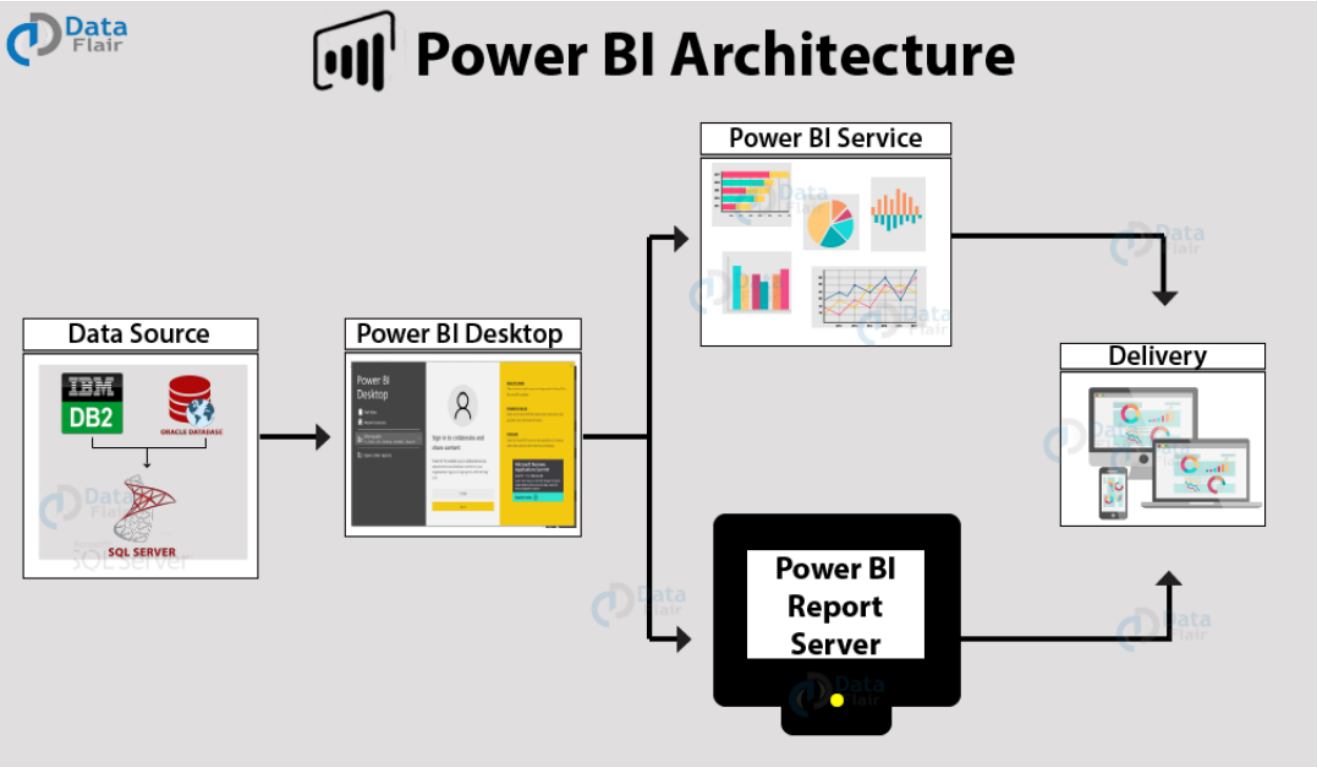
  "2024 Gartner Magic Quadrant for **Analytics**

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POWER BI ARCHITECTURE

- Power BI gives the ability to analyze and explore data on premise as well as in the cloud.
- Power BI architecture is a service built on top of Azure. There are multiple data sources that Power BI can connect to.



POWER BI ARCHITECTURE

- **Power BI desktop** is a component primarily used for Business reports generation and desktop creation. Also, all the other works typically begin with Power BI desktop, where the report creation takes place.
- **Power BI services** refer to the cloud services that are used to publish Power BI reports and data visualizations. Power BI service is the Software as a Service (SaaS) part of Power BI.
- Using **Power BI mobile apps**, you can stay connected to their data from anywhere. Power BI apps are available for Windows, iOS, and Android platforms.

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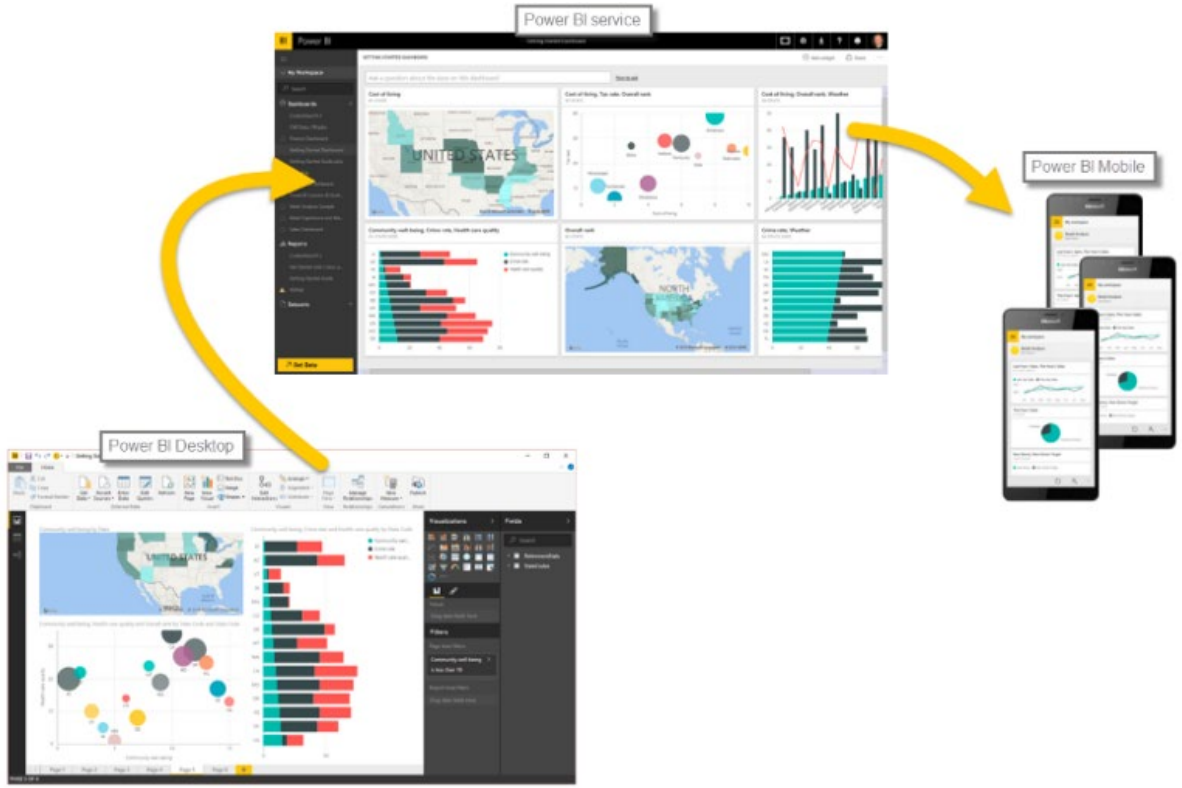
POWER BI OVERVIEW

- Power BI provides the ability to collaborate and share customized dashboards and interactive reports across colleagues and organizations, easily and securely.



POWER BI DESKTOP

- Usually, an analysis work begins in **Power BI Desktop** where report creation takes place.
- The report is then published to **Power BI service** from where it can be shared to the **Power BI Mobile apps** so that people ca view the reports even on mobiles.

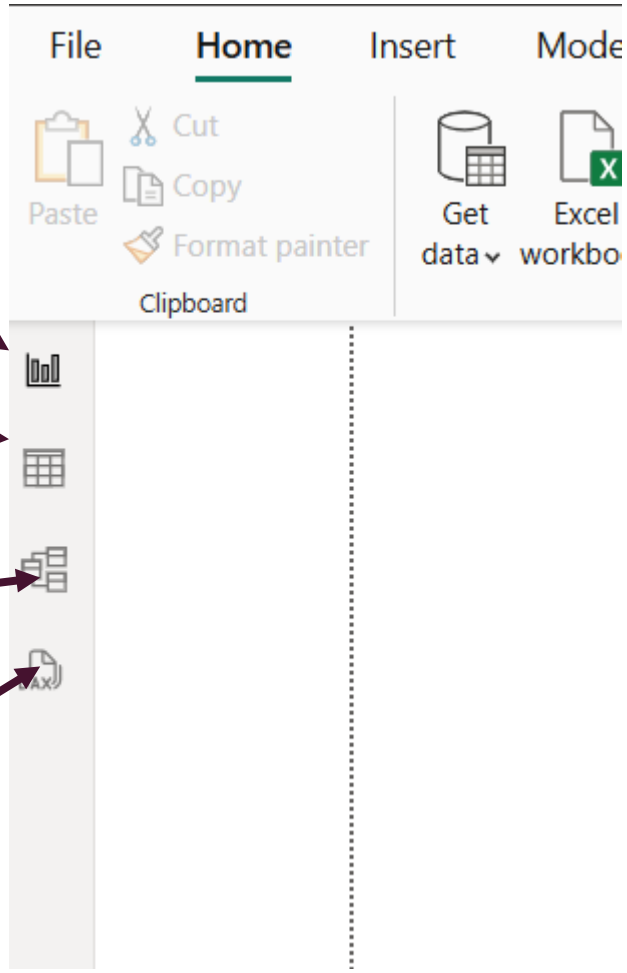


POWER BI DESKTOP

- Power BI Destop is a free application that can be downloaded and installed on the system.
- The most common uses for Power BI Desktop are as follows:
  -Connect to multiple data sources
  -Transform and clean that data, to create a data model
  -Create visuals, such as charts or graphs, that provide visual representations of the data
  -Create reports that are collections of visuals, on one or more report pages
  -Share reports with others by using the Power BI service

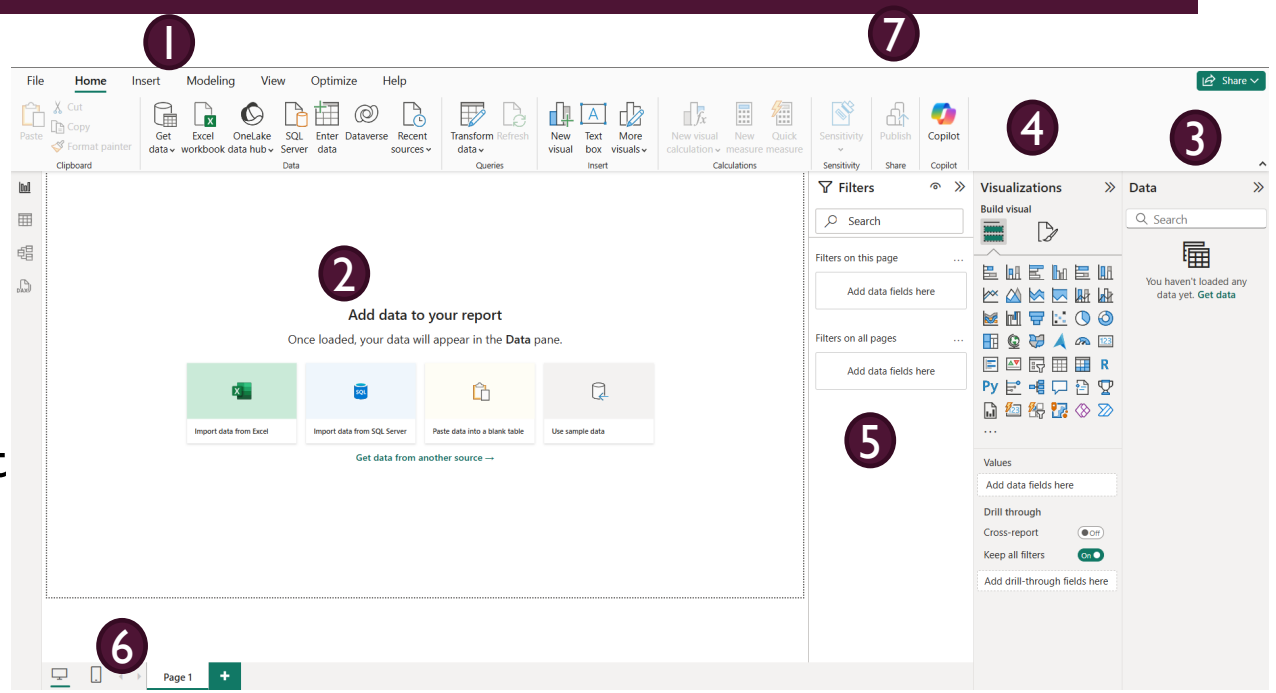
POWER BI DESKTOP WORKSPACE

- **Basic Views**
- **-Report View**: In this view, you create reports and visuals, where most of your creation time is spent.
  -Table View: The table view gives a preview of the entire data. Tables, measures, and other data used in the data model associated with the report, and transform the data for best use in the report's model
  -Model View: The model view displays the relationship between various objects.
- **-DAX query view-** view and work with Data Analysis Expressions (DAX) queries on semantic models.



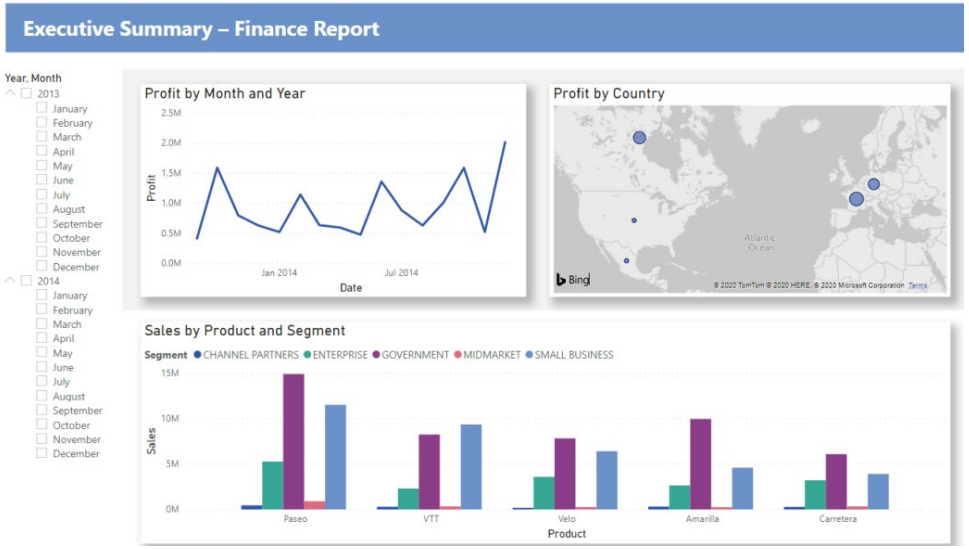
REPORT VIEW

- **1.Get Data**: allows you to import your data in the form of excel files, text/csv, xml, etc. We can also obtain data from Microsoft Azure, or other online services like google analytics, GitHub, etc.
  2.**Canvas** : where visualizations are created and arranged.
  3. **Datas:** Here you can view the data you import and it's fields.
  4.**Visualizations**: This pane consists of the different visuals that you can utilize to display distinctive visualizations of your data.
- 5. **The Filters pane**: where you can filter data visualizations
- 6. **The pages tab** :which lets you select or add report pages
  7.**Publish**: enables you to publish your data model to your Power BI workspace.



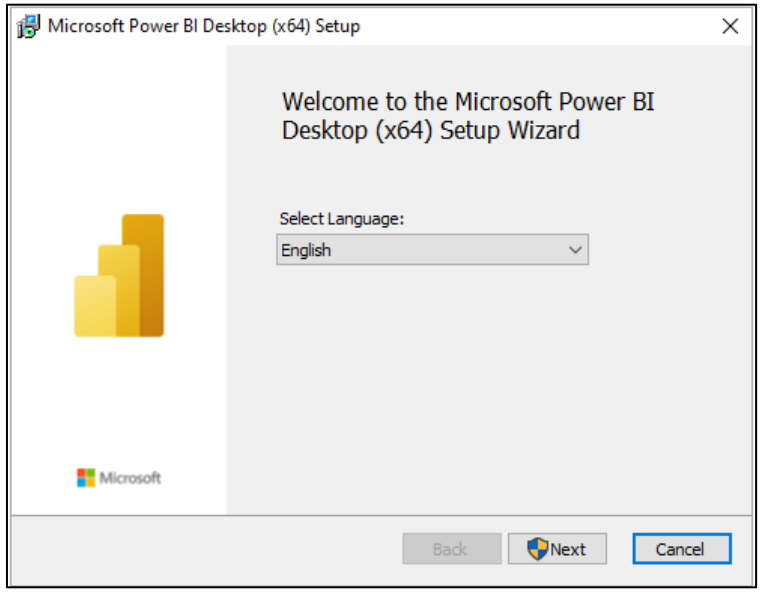
REPORTS

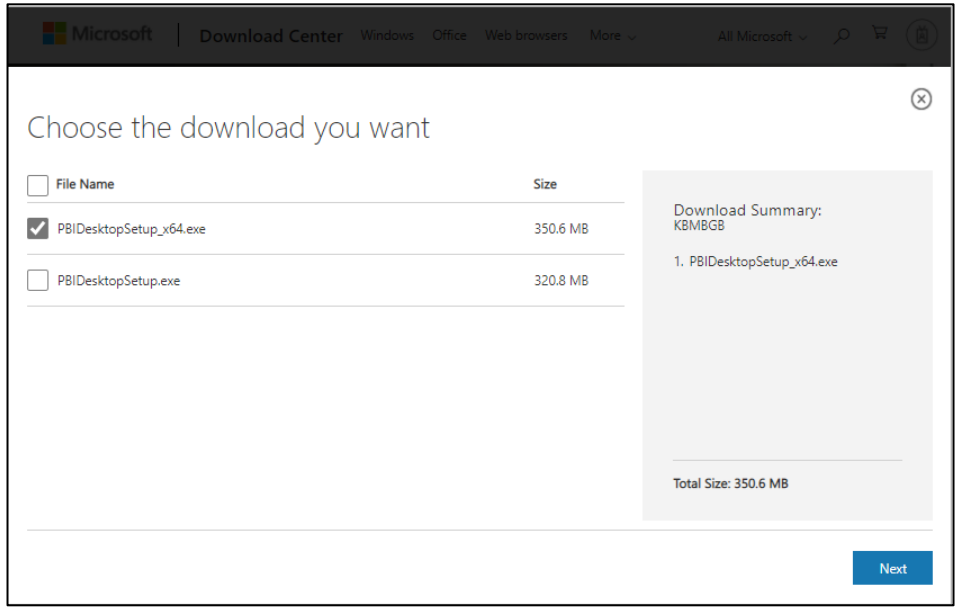
- A collection of visuals, in one Power BI Desktop file, is called a report.
- A report can have one or more pages, just like an Excel file can have one or more worksheets.
- With Power BI Desktop you can create complex and visually rich reports, using data from multiple sources, all in one report that you can share with others in your organization.



INSTALLATION

- To download the Power BI Desktop executable from the Download Center, select **Download** from the Download Center page (https://www.microsoft.com/download/details.aspx?id=58494).
- Specify a 32-bit or 64-bit installation file to download.
- Follow through the installation wizard until finish.

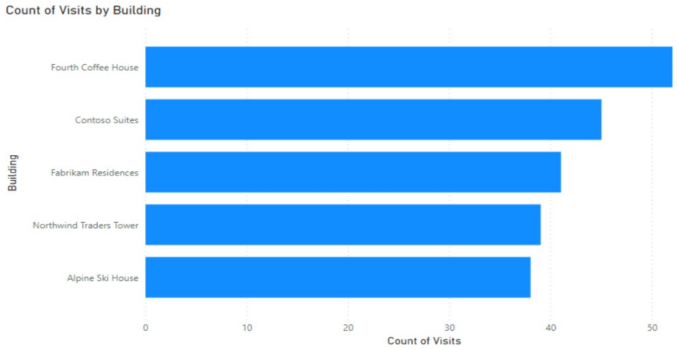


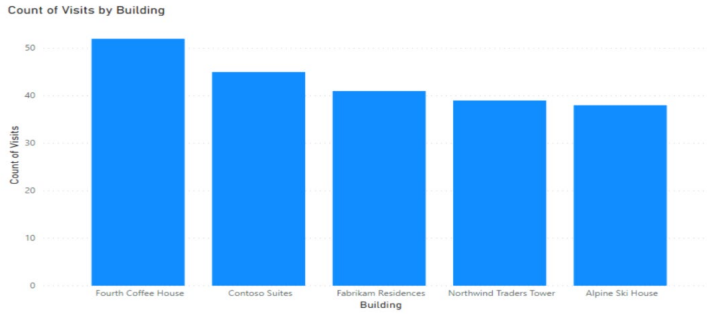


TYPES OF VISUALISATIONS

1. Bar and column charts

Bar charts are the standard for looking at a specific value across different categories.





2. Cards: Single number



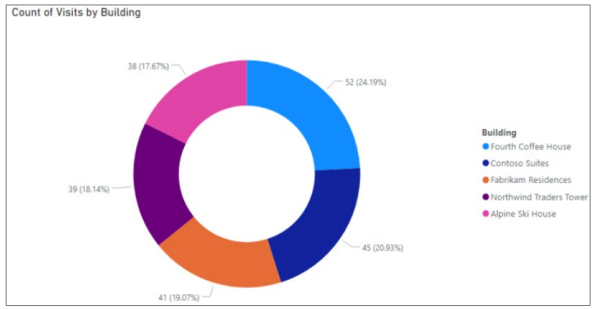
3. Cards: Multi row



TYPES OF VISUALISATIONS

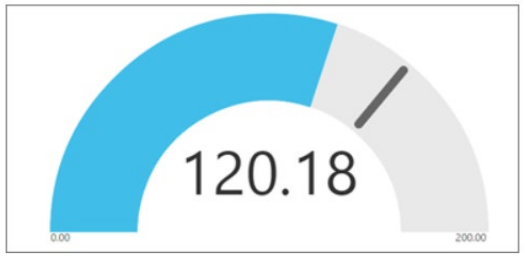
4. Donut charts

Donut charts are similar to pie charts. They show the relationship of parts to a whole.



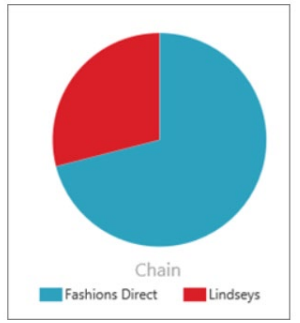
6. Gauge charts

Displays current status in the context of a goal.



5. Pie charts

Pie charts show the relationship of parts to a whole.



7. Maps: Basic maps

Used to associate both categorical and quantitative information with spatial locations.



TYPES OF VISUALISATIONS

8. Matrix

A table supports two dimensions, but a matrix makes it easier to display data meaningfully across multiple dimensions -- it supports a stepped layout. The matrix automatically aggregates the data and enables drill down.

Quarter	Q1		Q2
Year	Revenue	YTD Revenue	Revenue
2010	$45,186,241.942528	45,186,241.94	$70,609,615.88247627
2011	$47,641,801.21002861	47,641,801.21	$71,129,455.27992704
2012	$53,054,664.287495166	53,054,664.29	$72,265,427.93995765
2013	$45,011,474.33002325	45,011,474.33	$74,627,866.02998048
2014	$45,648,089.42003315	45,648,089.42	$77,663,582.71497234
2015	$45,186,241.942538336	45,186,241.94	$70,609,615.88244378
Total	$281,728,513.13264656	45,186,241.94	$436,905,563.72975754

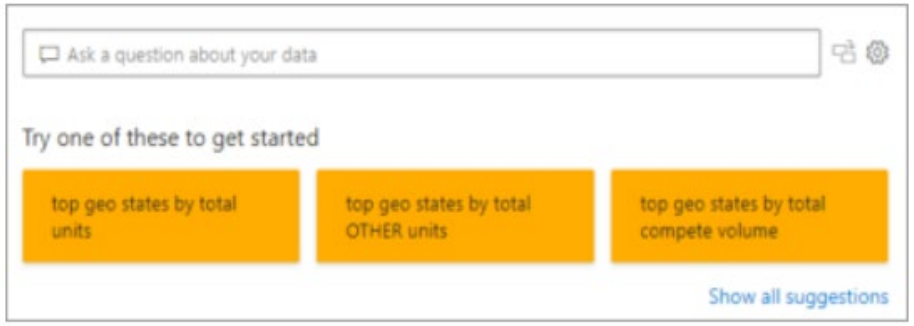
9. Tables

Works well with quantitative comparisons among items where there are many categories.

District	Count of StoreNumber	Average Selling Area Size
FD - 01	9	43,889
FD - 02	9	47,778
FD - 03	9	50,000
FD - 04	10	50,500
LI - 01	13	10,385
LI - 02	11	10,909
Total	**61**	**33,361**

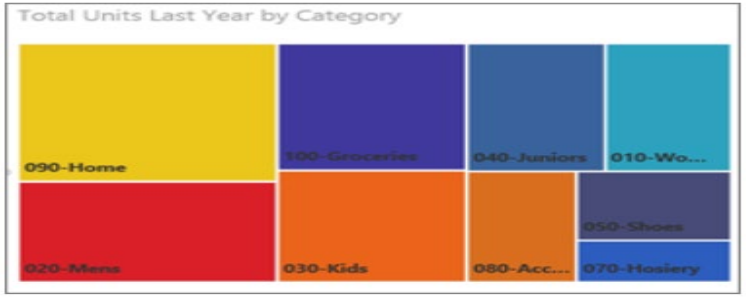
10. Q&A visual

The Q&A visual lets you ask questions about your data using natural language.



11. Treemaps

Are charts of colored rectangles, with size representing value. They can be hierarchical, with rectangles nested within the main rectangles.

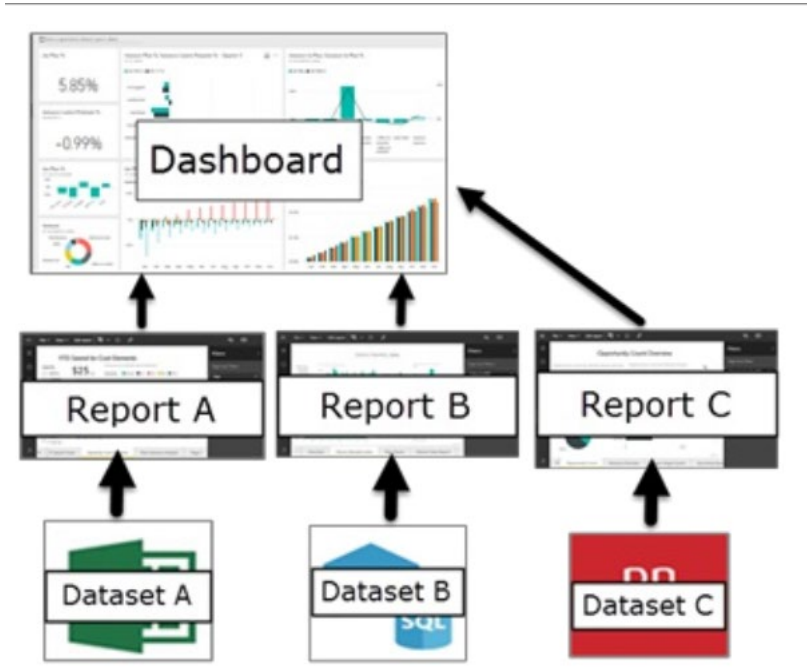


POWER BI SERVICE

- Power BI is a collection of software services, apps, and connectors that work together to create, share,and consume business insights in the way that serves you and your business most effectively.
- The Microsoft Power BI service (app.powerbi.com), sometimes referred to as Power BI online, is the SaaS (Software as a Service) part of Power BI.
- In the Power BI service, dashboards help to keep a finger on the pulse of the business. Dashboards display tiles, which user can select to open reports for exploring further.
- Dashboards and reports connect to datasets that bring all of the relevant data together in one place

DASHBOARD

- A Power BI dashboard is a single page, often called a canvas, that tells a story through visualizations. Because it's limited to one page, a well-designed dashboard contains only the highlights of that story. Readers can view related reports for the details.
- Dashboards are a feature of the Power BI service only. They're not available in Power BI Desktop. Although you can't create dashboards on mobile devices, you can view and share them there
- The visualizations you see on the dashboard are called tiles. You pin tiles to a dashboard from reports.
- The visualizations on a dashboard originate from reports and each report is based on a dataset. Selecting a visualization takes you to the report (and dataset) that it's based on



DATA SET

- The dataset you will you use today is a sales and market share analysis. This type of analysis is very common for the office of a Chief Marketing Officer (CMO). Unlike the office of the Chief Financial Officer (CFO), a CMO is focused not only on company's performance internally (how well do our products sell) but also externally (how well do we do against the competing products).
- The company, VanArsdel, manufactures expensive electronic products that could be used for fun as well as work and it sells them directly to consumers nationwide as well as several other countries. VanArsdel and its competitors have retained a 3rd party marketing company to collect and anonymize industry sales so that all participants can benchmark themselves.
- Download the dataset (USSales and Internatioal sales and copy all files in folder name DIAD/Data

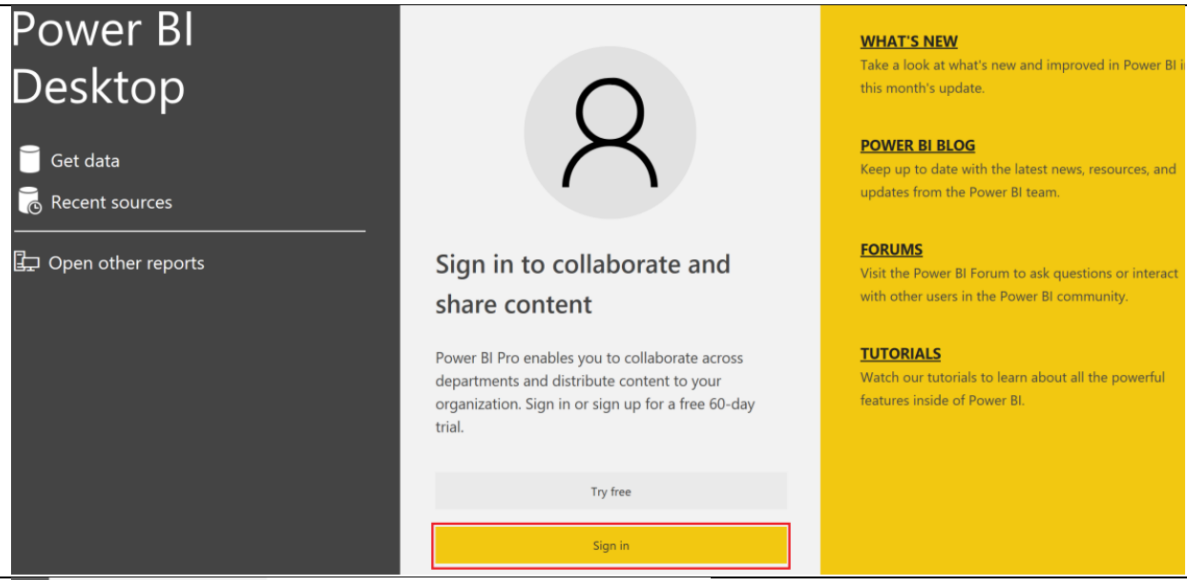
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DATA PREPARATION

POWER BI DESKTOP - LOAD DATA

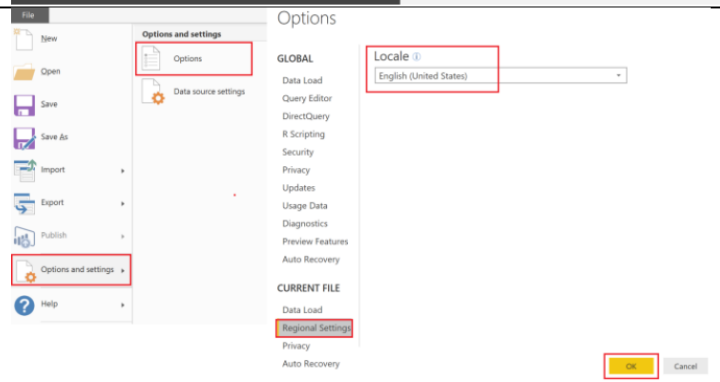
Steps

1. If you don't have the **Power BI Desktop** open, launch it now.
2. Startup screen opens. Click on **Sign In** and sign in using your Power BI credentials. Signing in to Power BI Desktop helps later when you are publishing to Power BI Service.



Let's set up the locale to US English, to make it convenient to go through the rest of this lab.

3. From the ribbon, select **File -> Options and settings -> Options**.
4. In the left panel of Options dialog, select **Regional Settings**.
5. From the **Locale** drop down select **English (United States).**

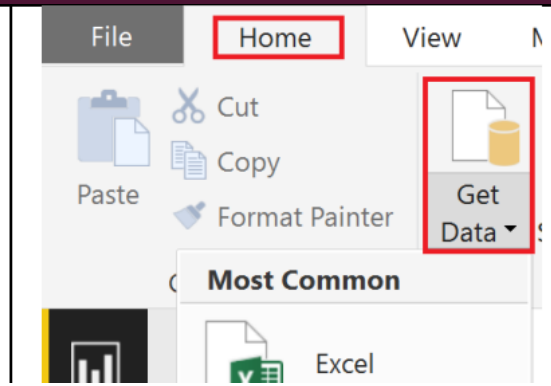


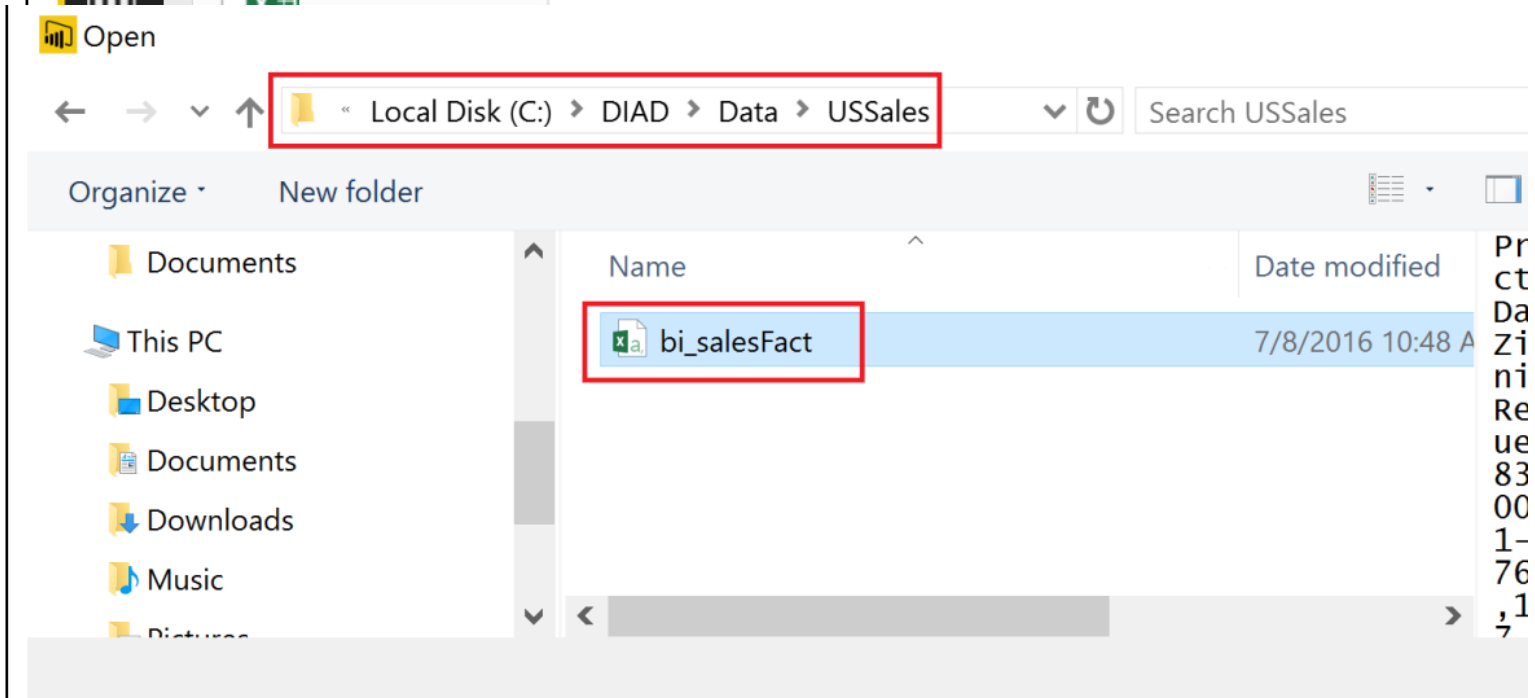
POWER BI DESKTOP - LOAD DATA

First step is to load data to Power BI Desktop. We will load USA Sales data which is in CSV files.

6. From the ribbon, select **Home -> Get Data.**
7. Select **Text/CSV.**

8. Browse to **DIAD\Data\USSales** folder and select **bi_salesFact.csv.**
9. Click **Open**.

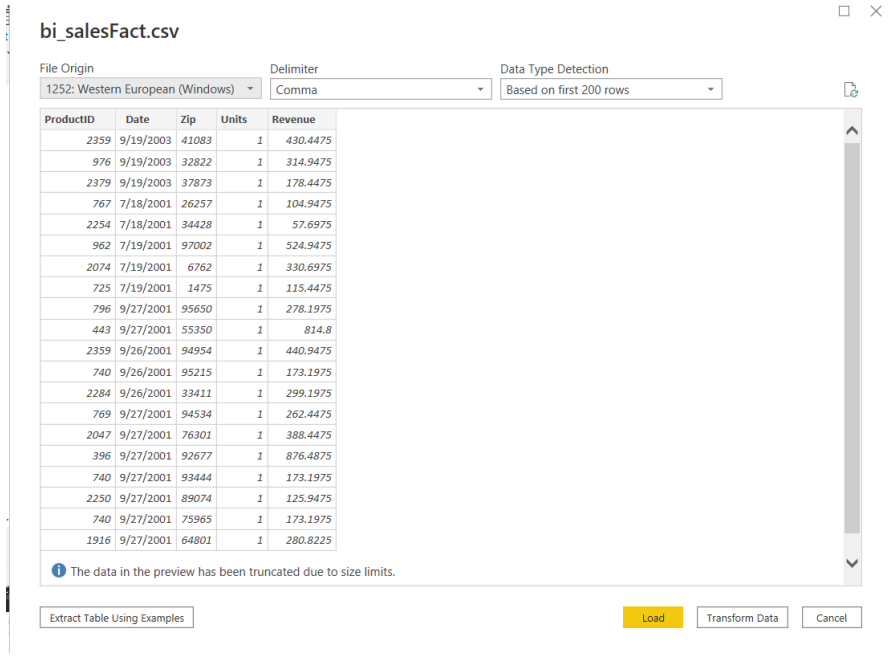


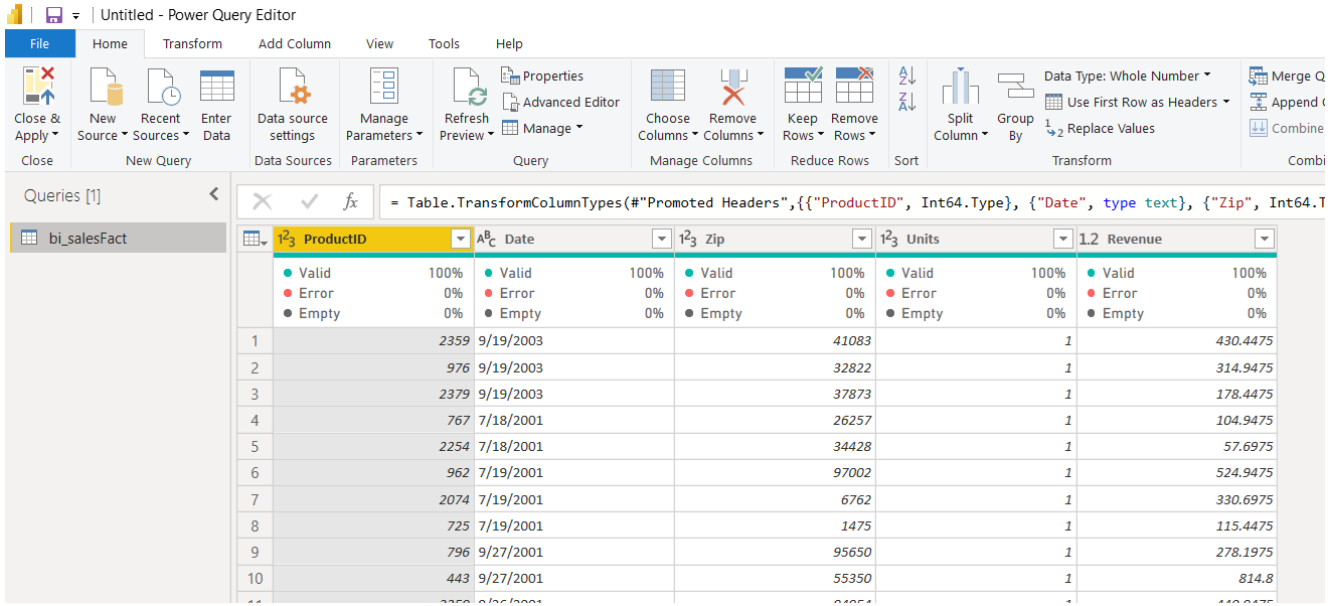


POWER BI DESKTOP - LOAD DATA

- After completing your selection, you have three options – Load, Transform data or Cancel. Click Transform data.
- Now you have loaded the Sales data into the file. You should be in the Query Editor window as shown in the picture to the right . Notice data type of each field is indicated next to the column header

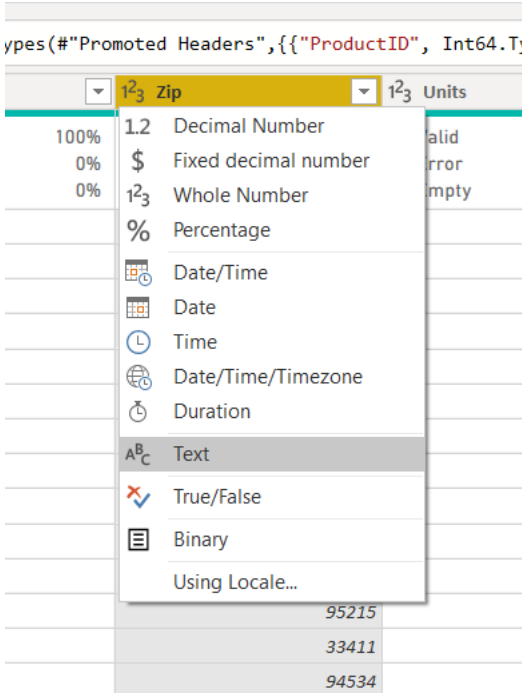
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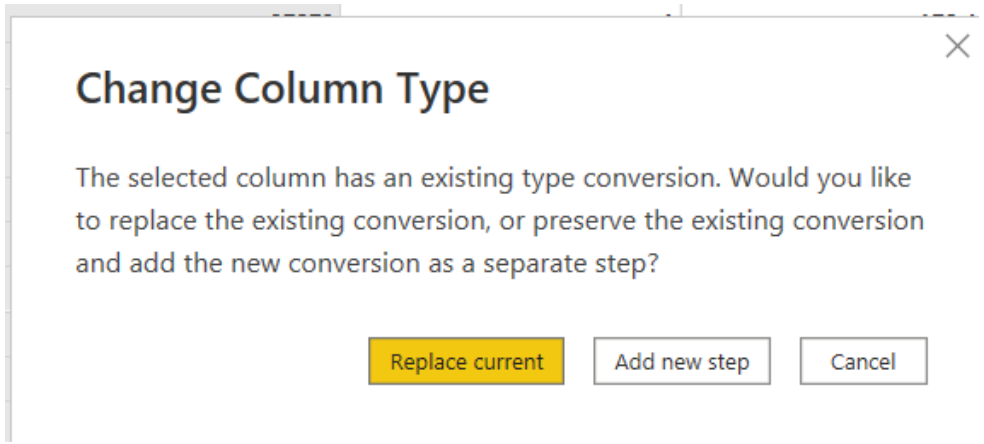




POWER BI DESKTOP - LOAD DATA

- Notice Power BI has set Zip field to data type Whole Number. To ensure that Zip codes which start with zero don't lose the leading zero, we will format them as text. Highlight the Zip column and change the Data Type to Text.
- Change Column Type dialog opens. Select Replace Current button.

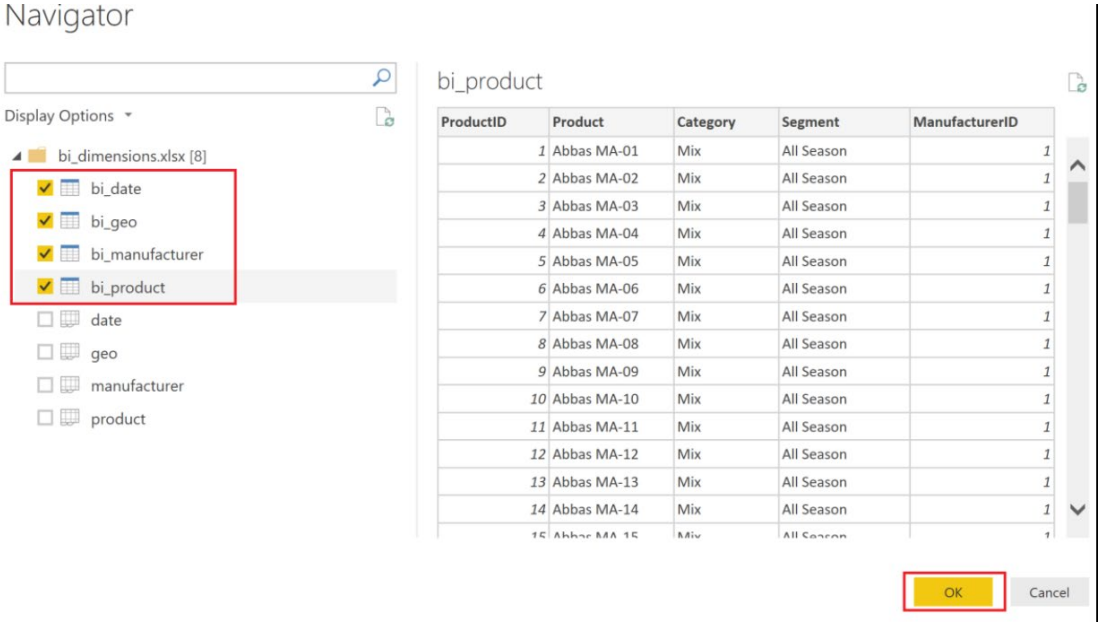




POWER BI DESKTOP - LOAD DATA

- Now let's get the data that is in excel source file.

1. From the ribbon, select Home -> New Source -> Excel
2. Browse to DIAD\Data\USSales folder and select bi_dimensions.xlsx.
3. In the Navigator window select all the tables. Select each table to preview the columns and rows. Note: The first four items are the named Excel Tables and the second set are the Excel Worksheet names. Table names are differentiated from Worksheet names by using different icons.
4. Click OK to edit these tables in the query editor. Notice 4 new queries are added to the query editor.



POWER BI DESKTOP - LOAD DATA

- To analyze all the data together you will want to import the new data from each of the subsidiaries and combine it with the US Sales you loaded earlier.

1. Click on the New Source drop down in the Home menu tab of the Query Editor.
2. Select More… as shown in the figure. Get Data dialog opens.. In the Get Data dialog select Folder. Click Connect
3. Folder dialog opens. Click Browse… button. In the Browse for Folder dialog navigate to the InternationalSales folder.
4. Click OK (to close the Browse for Folder dialog box) and Click OK (to close the Folder dialog box).
5. Since we want to combine data, click Combine & Transform Data

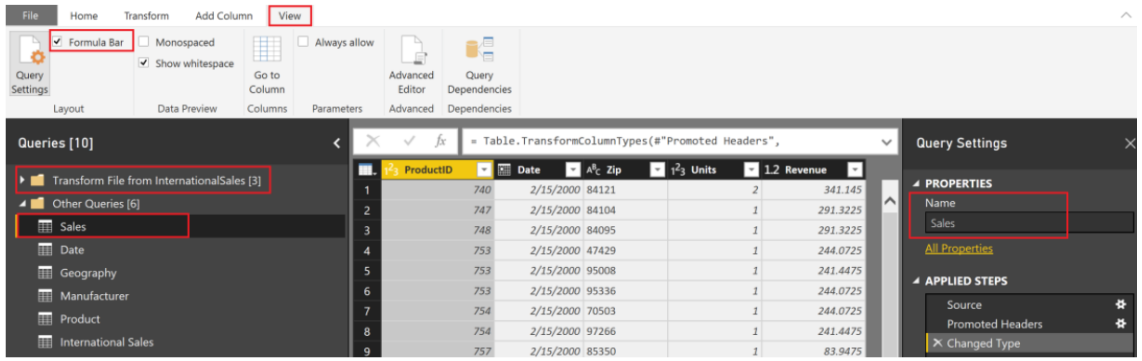
POWER BI DESKTOP - TRANSFORMING YOUR DATA

The Query Editor window should appear as shown in the diagram.

- If formula bar is disabled, you can turn on the formula bar from the View ribbon. This enables you to see the “M” code generated by each click on the ribbons.
- Select the options available on the ribbon – **Home, Transform, Add Column and View** to notice the various features available.

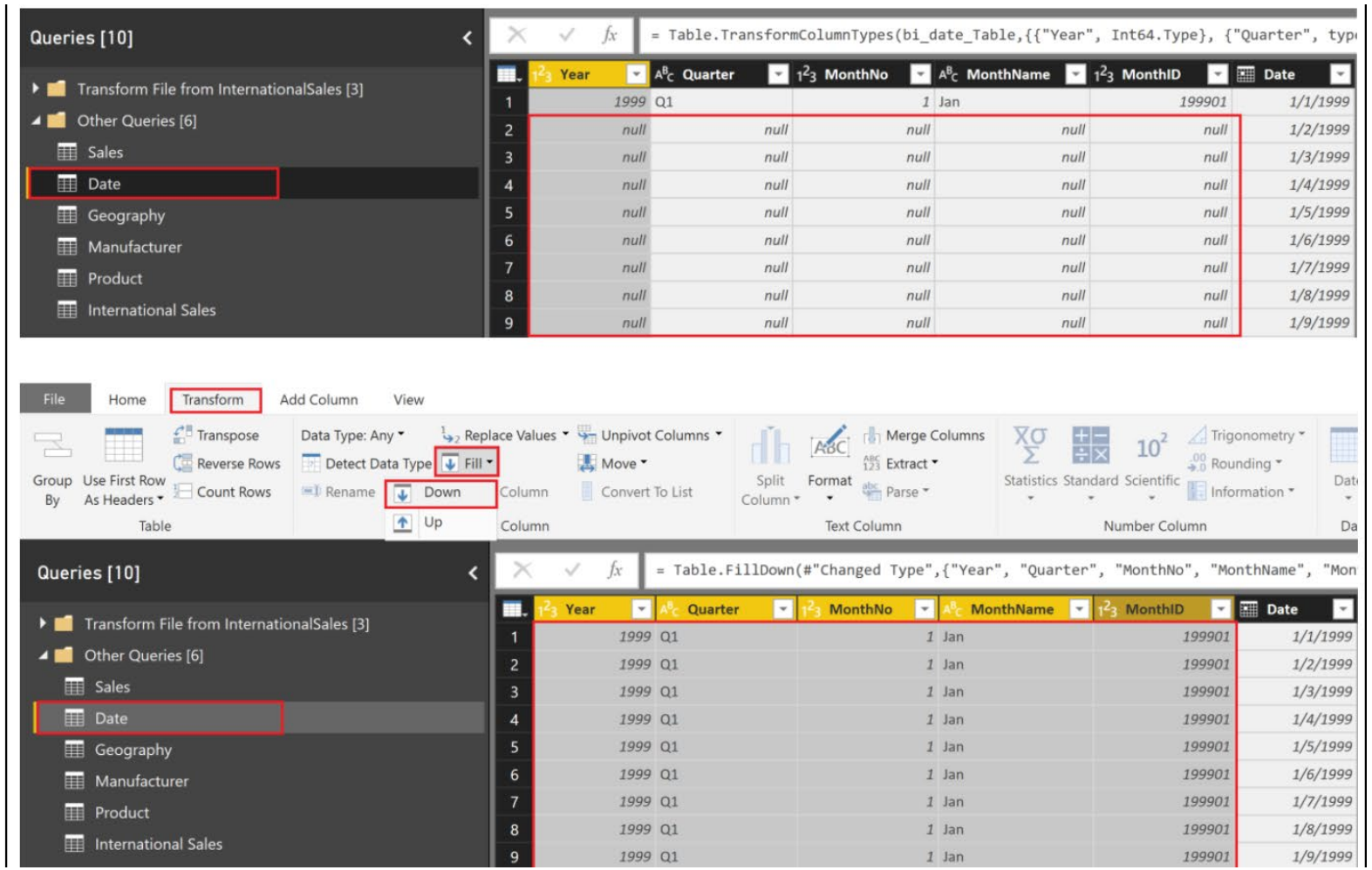
1. Under **Queries** panel**, minimize** Transform Binary from InternationalSales folder.
2. Select each query name in the **Other Queries** section**.**
3. **Rename** them in the Query Settings -> Properties section as shown below:

Initial Name	Final Name
bi_salesFact	Sales
bi_date	Date
bi_geo	Geography
bi_manufacturer	Manufacturer
bi_product	Product
InternationalSales	International Sales



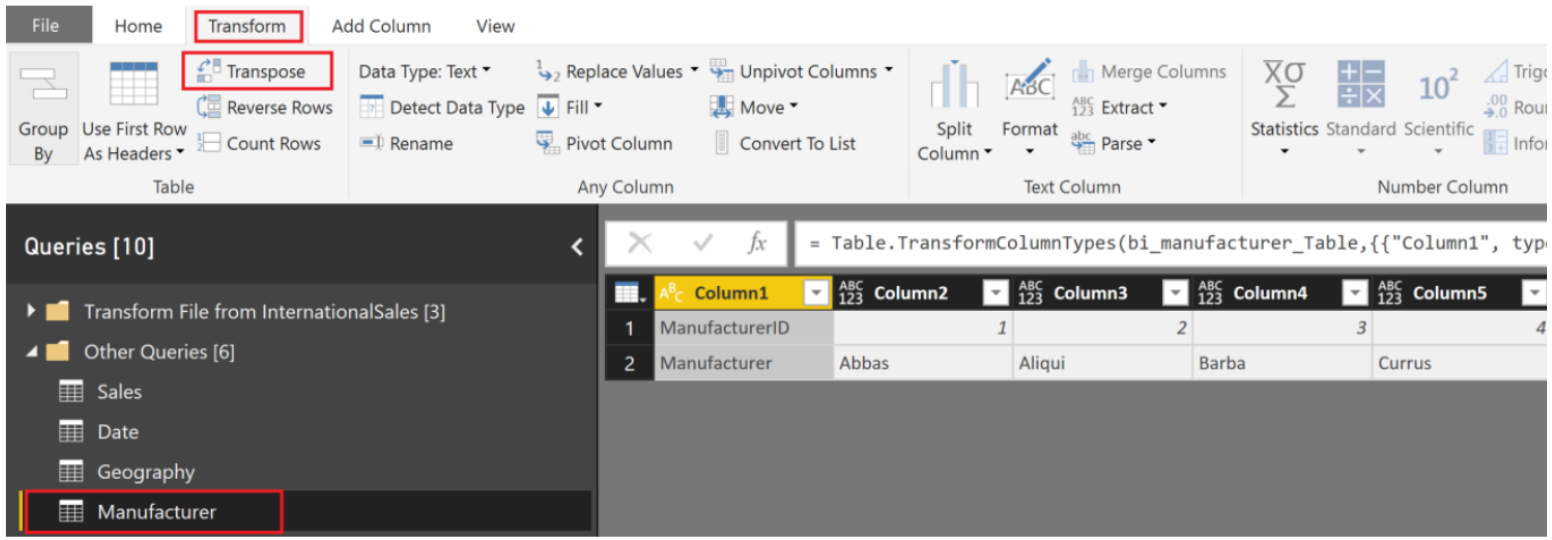
4. From the left panel, select **Date** Query.
5. Using Ctrl left click key select **Year, Quarter, MonthNo, MonthName and MonthID** fields.
6. From the ribbon select **Transform -> Fill -> Down**.

Notice now all the null values are filled with the appropriate Year, Quarter, MonthNo, MonthName, MonthID values.



7. From the left panel, select **Manufacturer** Query. Notice ManufacturerID and Manufacturer data is laid across in rows. And the header is not useful. We need to transpose the table to meet our needs.
8. From the ribbon select **Transform -> Transpose**.

Notice this transposes the data into columns. Now we need the first row to be the header.



9. From the ribbon select **Home -> Use First Row As Headers**.

Notice now Manufacturer table is laid out the way we need it with a header and values along columns.
Also, notice in the **APPLIED STEPS** section on the right panel, all the steps that we add is being recorded.

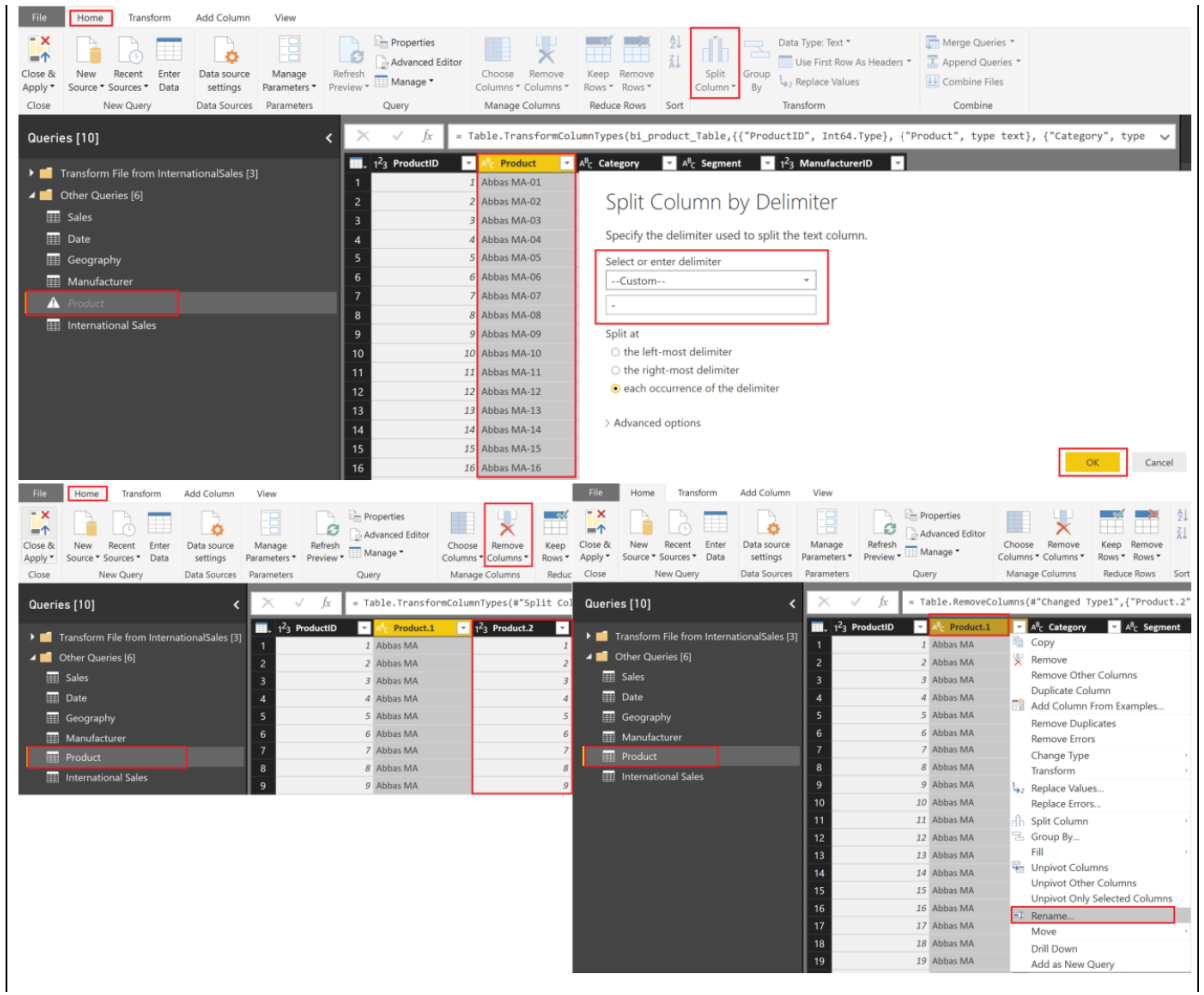
10. From the left panel, select **Product** Query. Notice Product field is a concatenation of Product Name and Product ID. For our scenario, we need only the Product Name.
11. Select **Product column**.
12. From the ribbon select **Home -> Split Column -> By Delimiter**.
13. Dialog opens. From the dialog, select **–Custom—**from the dropdown.
14. Enter "**-**" in the text area since we want to split the column based on the hypen.
15. Select **OK.**

Notice Product column is split into two columns, Product.1 and Product.2.
We do not need Product.2 column since it is the Product ID, which is already available.

16. Select **Product.2 column**.
17. From the ribbon select **Home -> Remove Columns** to remove the column.

Let's rename Product.1 column to Product.

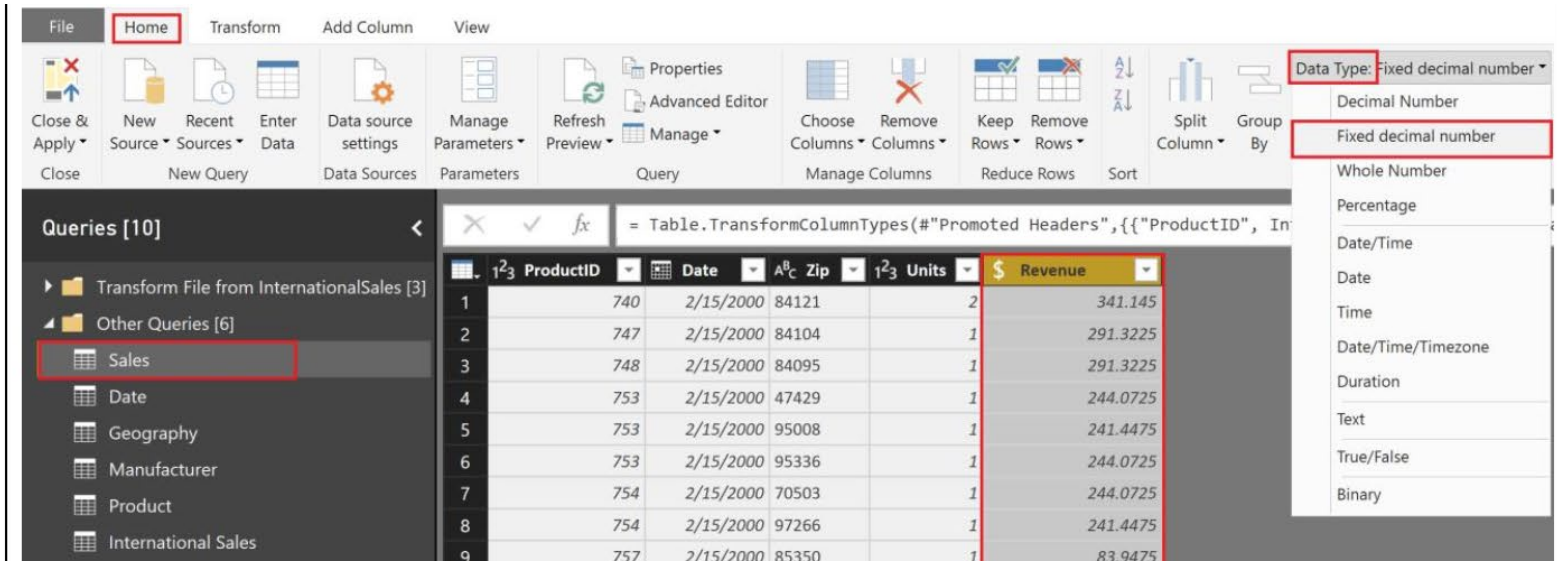
18. Right click on **Product.1 column** header.
19. From the menu select **Rename** to rename the column to **Product**.



20. From the left panel, select **Sales** Query.
21. Verify and if needed, **Replace Current** Data Type using **Home -> Data Type** drop down as shown below:

Column	Data Type
Date	Date
Revenue	Fixed Decimal Number

Select **Replace Current** in the Change Column Type dialog.

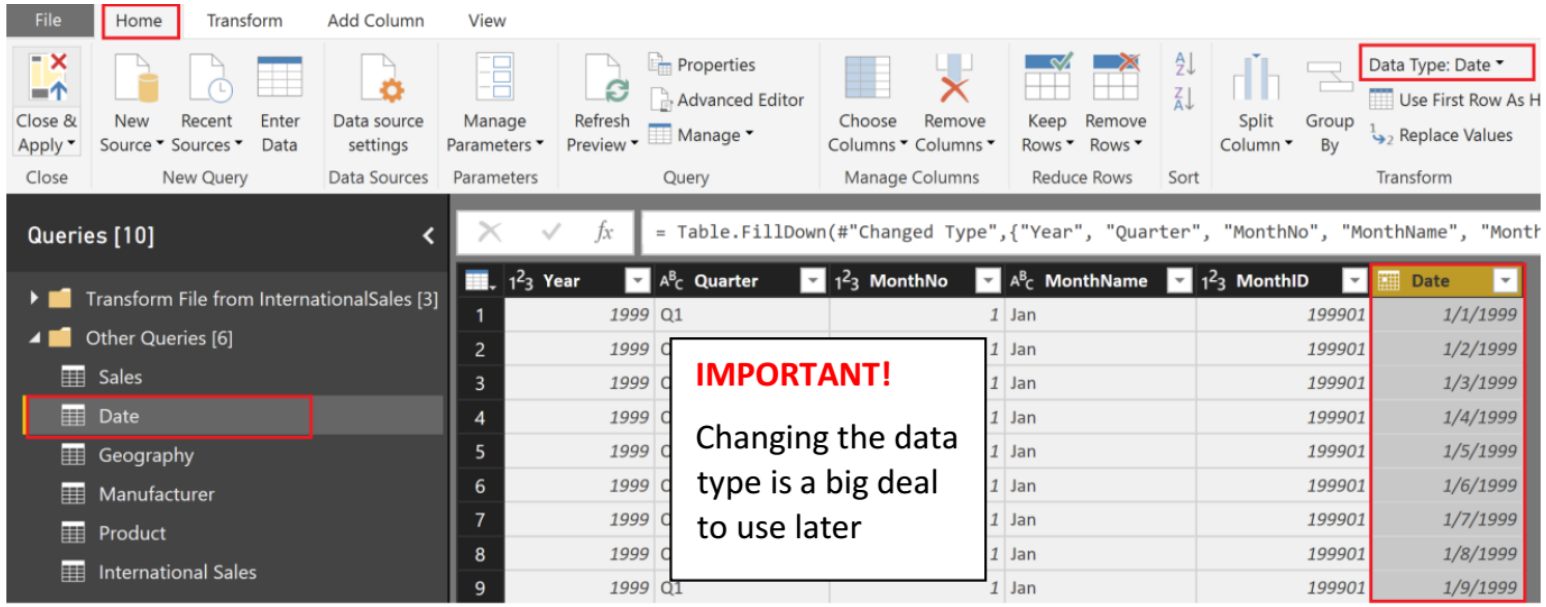


22. From the left panel, select **Date** Query.
23. Verify and if needed **Replace Current** Data Type using **Home -> Data Type** drop down as shown below.

Column	Data Type
Date	Date

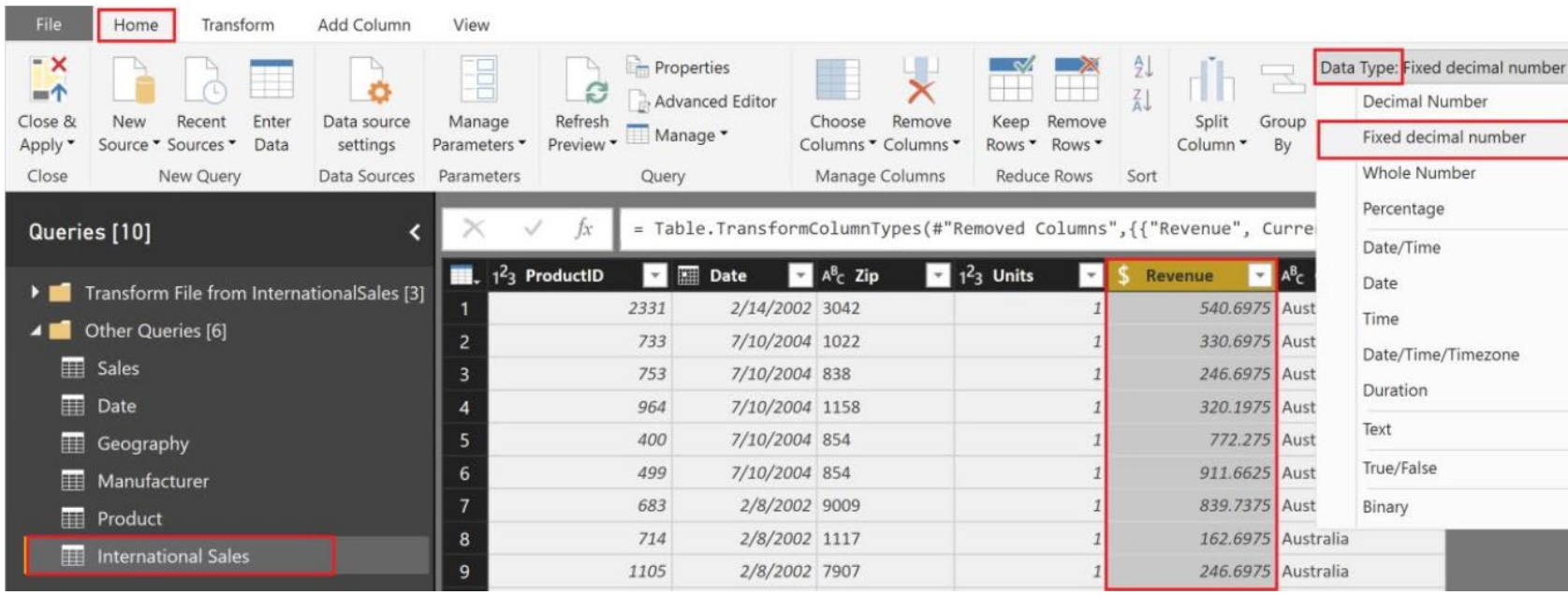
24. From the left panel, select **Manufacturer** Query.
25. Verify and if needed set Data Type using **Home -> Data Type** drop down as shown below.

Column	Data Type
ManufacturerID	Whole Number
ManufacturerName	Text

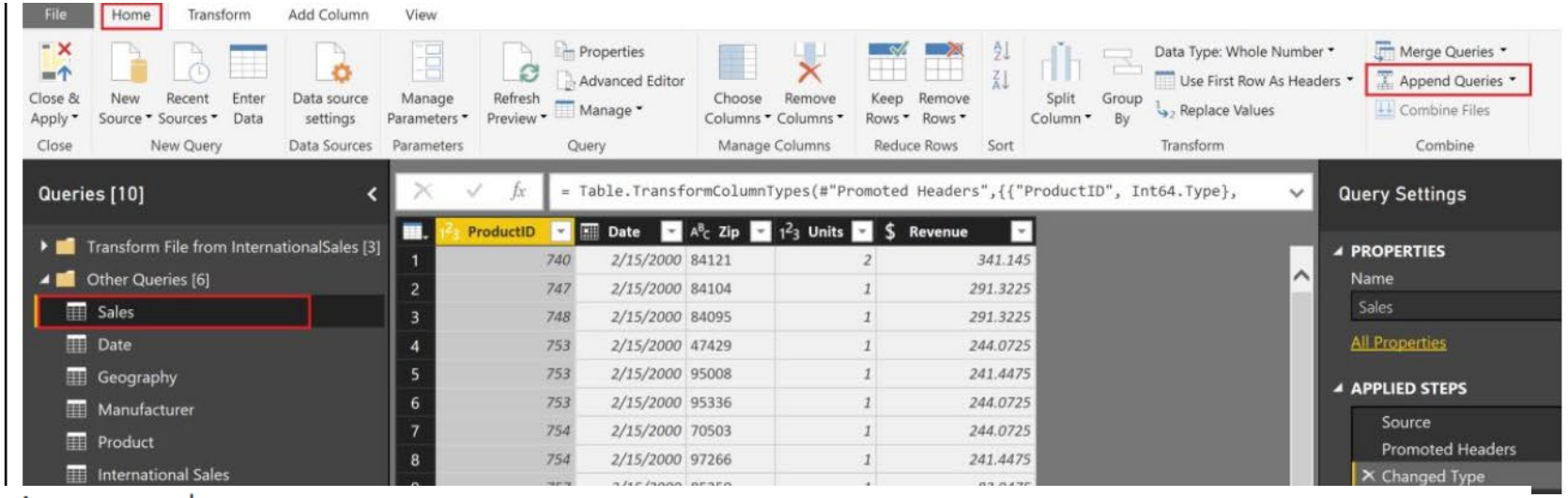


26. From the left panel, select **International Sales** Query.
27. Verify and if needed set Data Type using **Home -> Data Type** drop down as shown below.

Column	Data Type
ProductID	Whole Number
Date	Date
Zip	Text
Units	Whole Number
Revenue	Fixed Decimal Number



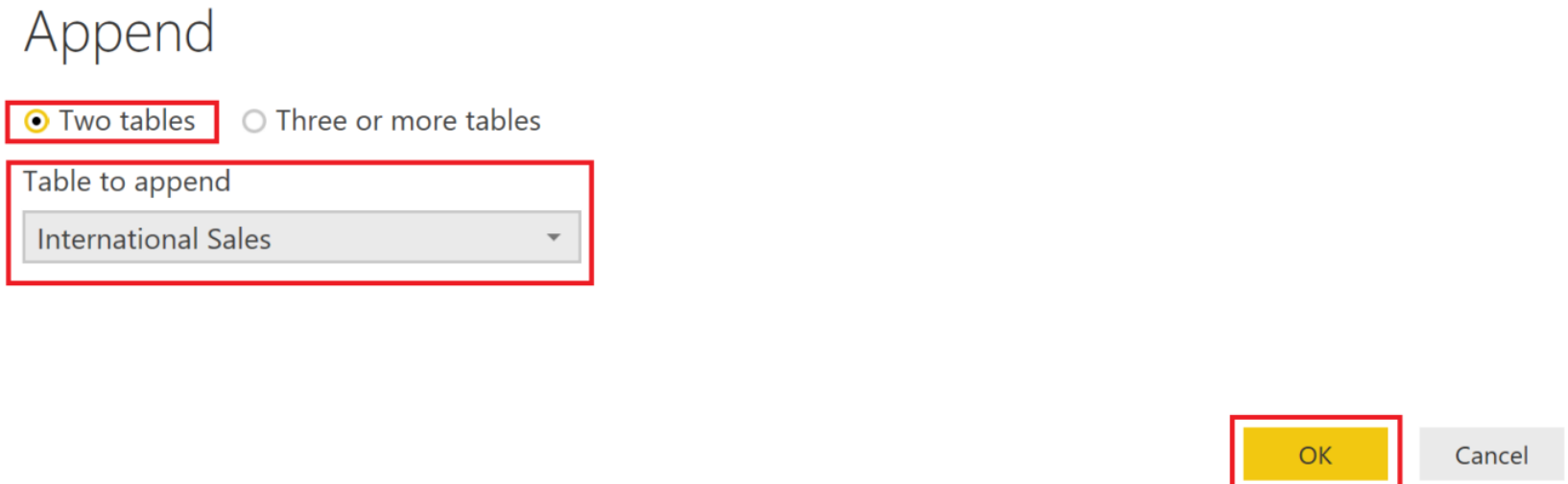
To analyze the Sales of all countries, it is convenient to have a single Sales table. Hence you want to append all the rows from **International Sales** to **Sales**.



28. Select **Sales** in the Queries window in the left panel as shown in the figure.
29. From the ribbon select **Home -> Append Queries** as shown in the figure.

Append dialog opens. There is an option to append **Two tables or Three or more tables**. Leave Two tables selected since we are appending just two tables.

30. Select **International Sales** from the drop down and click **OK**.

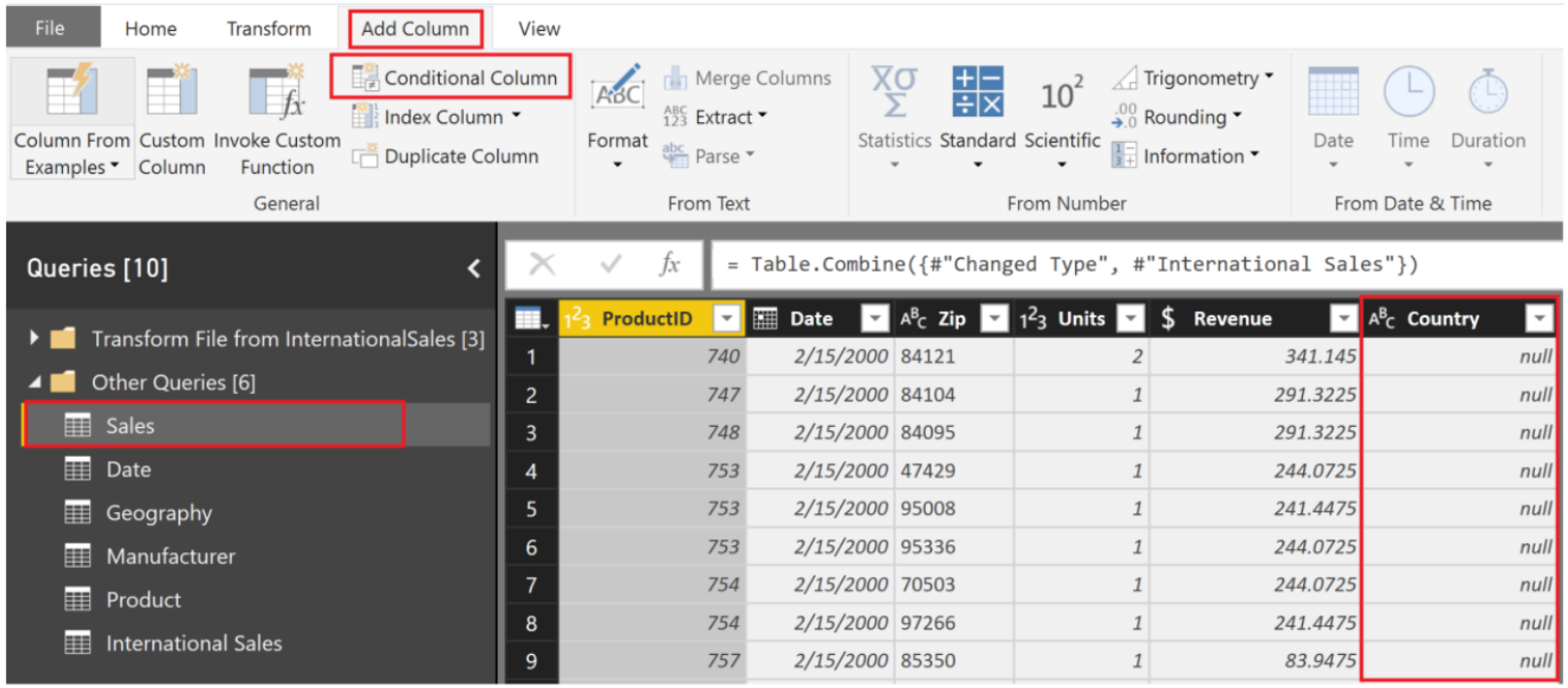


You will now see a new column in the **Sales** table called **Country**. Since International Sales had the additional column for Country, Power BI Desktop added the column to the Sales table when it loaded the values from International Sales.

You see **null values** in the **Country** column by default for the Sales table rows because the column did not exist for the table with USA data.

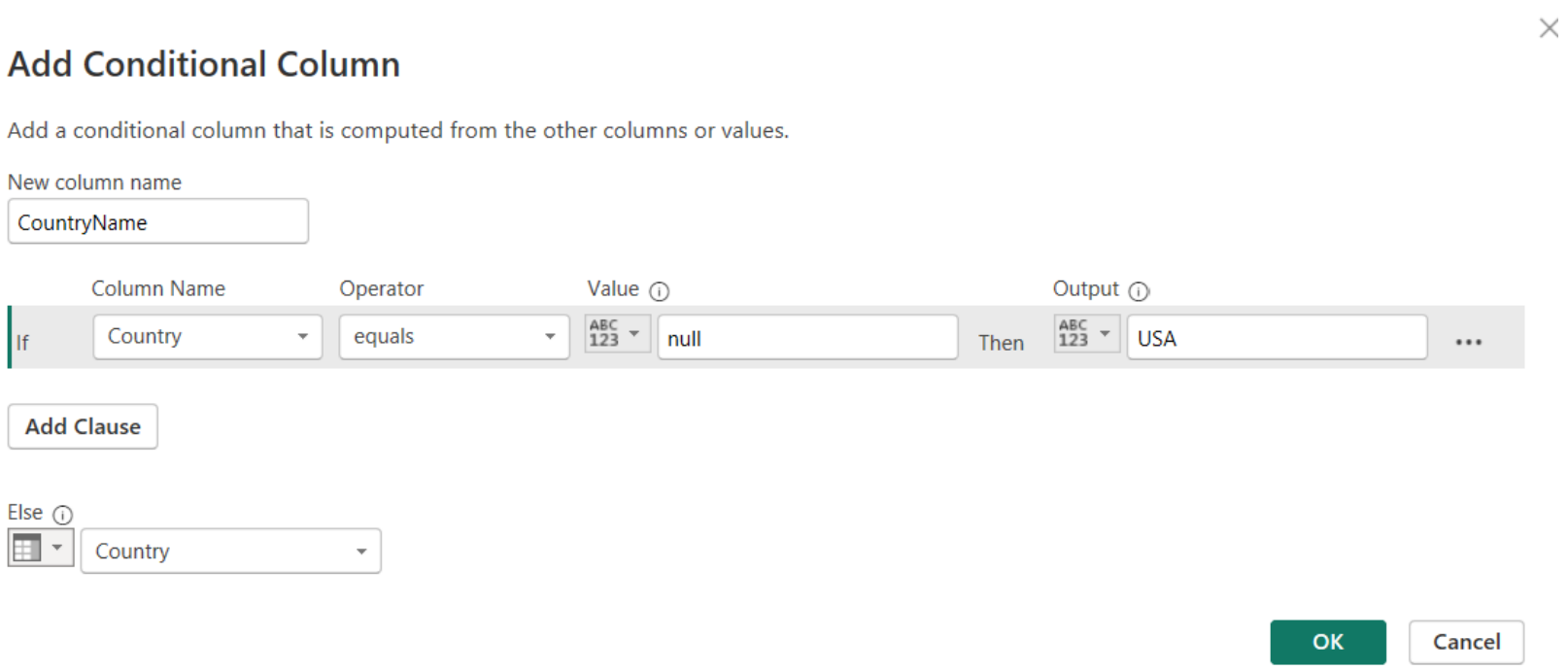
We will add the value "**USA**" as a data shaping operation.

31. From the ribbon select **Add Column -> Conditional Column** as shown in the figure.

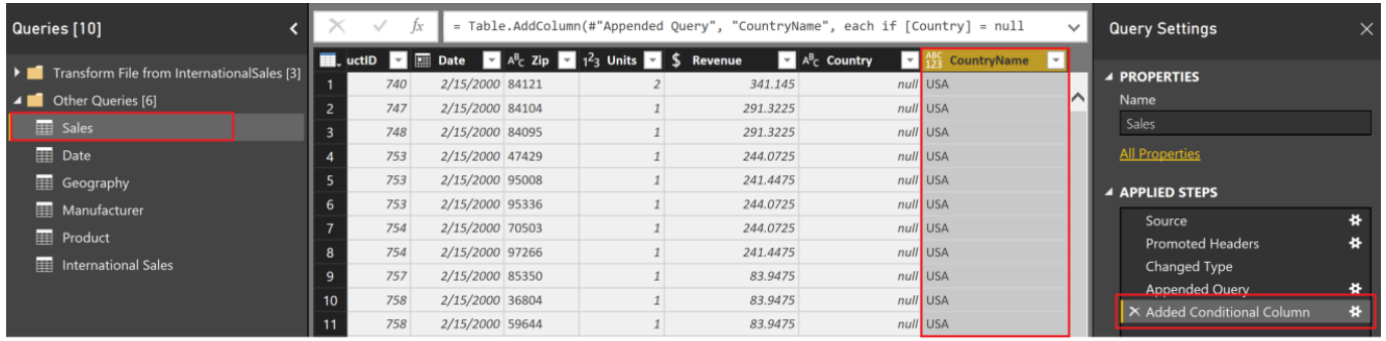


	ProductID	Date	Zip	Units	Revenue	Country
1	740	2/15/2000	84121	2	341.145	null
2	747	2/15/2000	84104	1	291.3225	null
3	748	2/15/2000	84095	1	291.3225	null
4	753	2/15/2000	47429	1	244.0725	null
5	753	2/15/2000	95008	1	241.4475	null
6	753	2/15/2000	95336	1	244.0725	null
7	754	2/15/2000	70503	1	244.0725	null
8	754	2/15/2000	97266	1	241.4475	null
9	757	2/15/2000	85350	1	83.9475	null

32. In the **Add Conditional Column** dialog, enter name of the column as "**CountryName**".
33. Select **Country** from the **Column Name** dropdown.
34. Select **equals** from the **Operator** dropdown.
35. Enter **null** in the **Values** text.
36. Enter **USA** in the **Output** text.
37. Select the dropdown under Else and pick **Select a column** option.
38. Select **Country** from the column dropdown.
39. Click **OK**.



40. You will see the **CountryName** column in the Query editor window.

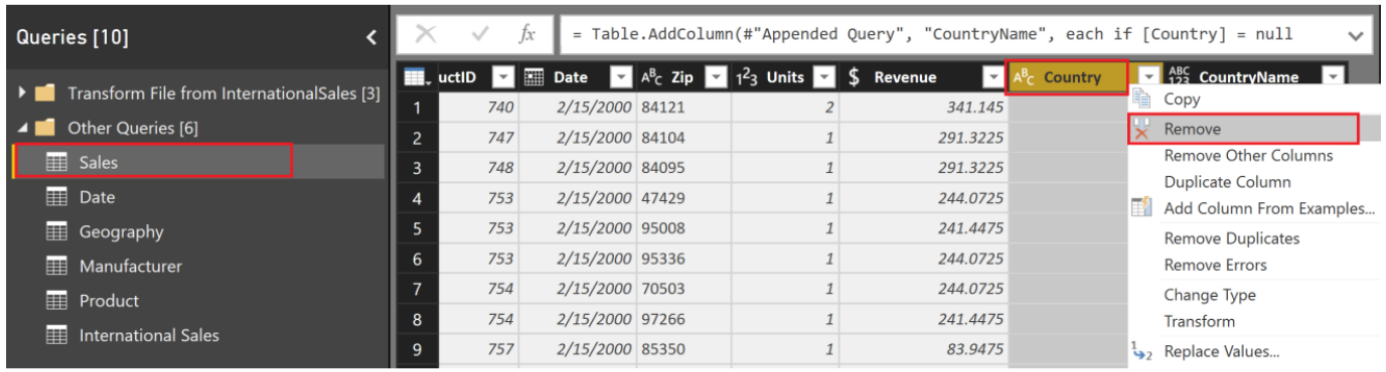


The original **Country** column is only required as a temporary column. It is not required in the final table for analysis and can be removed.

41. Right click on the **Country** column and select **Remove** as shown in the figure.

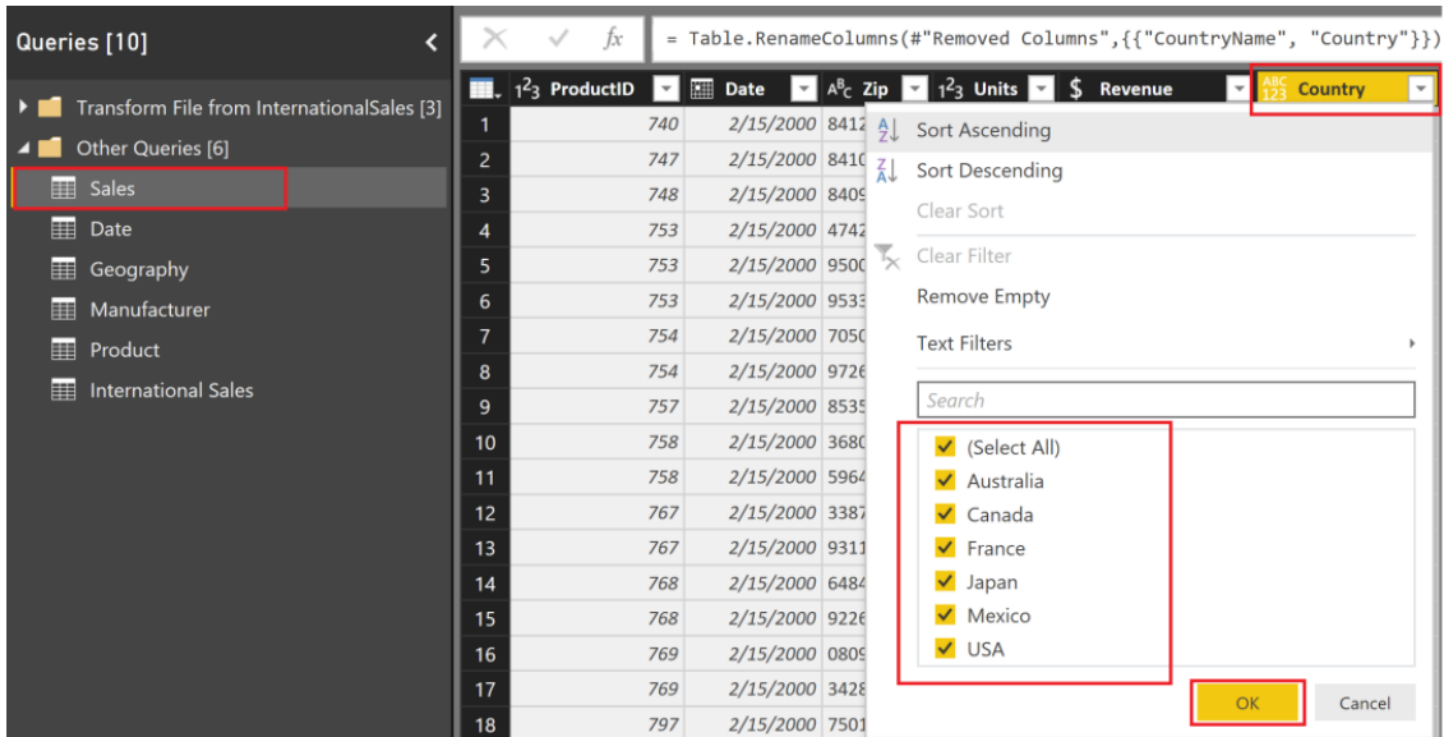
We can now rename **CountryName** column to Country.

42. Right click on the **CountryName** column and rename to **Country**.
43. Using **Home -> Data Type**, change the **data type** of the **Country** column to type **Text**.



The newly named **Country** column will have names for all countries, including the USA. You can validate this by clicking on the drop down next to **Country** column to see the unique values.

44. At first, you will only see USA data. Click on **Load more** to validate you have data from all 4 countries included.
45. Click **OK** to close this filter.



- We don't need the International Sales table to load to the data model now that its rows have been appended to the Sales table. Let's prevent the International Sales table from loading to the data model.
- From the Queries panel on the left, select International Sales query.
- Right click and select Enable Load. This will disable loading International Sales to prevent duplicate data.
- Select Home – Close and Apply

